Morphophonology

Topic 7: Morphology licensed by phonology?

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A modularity problem

- The application of phonology depends on morphological structure
  - Baseline assumption: all variation in the pronunciation of morphemes should depend on surrounding phonological context
  - Challenge: may depend on morphological context
- Recurring theme of this class: mechanisms for allowing phonology to be influenced by morphology
- So far: mostly, blocking morphology from applying in certain morphological contexts

Allomorphy

- An implicit goal: derive all variation in the pronunciation of morphemes from a single phonological grammar
  - We have seen some ways in which grammar may not apply uniformly in all morphological contexts
- What about cases where this fails?
  - "Allomorphy" (in the narrower sense)

Two important distinctions

- Can the alternation be derived with the phonological grammars?
- Can the contexts for alternation be derived with the phonological grammar

We have focused here

	In the expected contexts…		
Expected alternation…		Y	N
	Y	Phonology	Misapplication?
	N	Allomorphy?	Allomorphy

Easily "phonology"

- Catalan word-final devoicing
  - *casos* [kázus] ‘cases’ vs. *cas* [kás] ‘case’: one single UR: /kaz/
  - *lletosa* [ʎətózə] ‘milk-like.f’ vs. *lletós* [ʎətós]: one single UR: /-oz/
- Yiddish nasal place assimilation
  - 1/3pl suffix is -m̩ after labial stops (hɔbm̩ 'have'), -ŋ̍ after velar stops (zɔgŋ̍ 'say'), -n̩ elsewhere (vɪsn̩ 'know'): one single UR /n̩/

Maybe phonology?

- Korean conditional: -mjən after vowels, -ɨmjən after consonants
  - Looks like epenthesis, and ɨ is used for epenthesis in some contexts (loan adaptation)
  - Seen with numerous suffixes
  - But in other morphological contexts, epenthesis is not regularly employed (e.g., -nim)

Less and less like phonology

- Difficult phonology
  - English *sing* pronounced *sang* in the past tense
  - Lakhota 1sg wa-j… → bl…
  - Requires clever and abstract representations
- Probably not phonology, but sensible contexts
  - Korean nominative is -ka after vowels, -i after consonants
- Unlikely to be (entirely) phonology
  - German plurals: *Tür*[ən], *Tag*[ə], *Kind*[ɐ], *Auto*[s] (though see Trommer 2021)

Not phonology, but phonologically conditioned: Girona Catalan personal articles

- Tarragona: no allomorphy (< Latin 'illum'). Same as definite article

	__C	__V
Masc.	el Pere	l'Enric
Fem.	la Maria	l'Agnès

- Balearic: no allomorphy (< Latin 'domine')

	__C	__V
Masc.	en Pere	n'Enric
Fem.	na Maria	n'Agnès

- Girona: allomorphy (mixed system)

	__C	__V
Masc.	en Pere	l'Enric
Fem.	la Maria	l'Agnès

The problem of phonologically-conditioned allomorphy

- Phonologically-conditioned allomorphy (PCA)
- Allomorphs are not derivable with phonology, but their distribution is
- Should we let the phonology do the job of determining their distribution?
- We show here how this can be done
  - An important precursor to deciding whether it *should* be done this way…

Moroccan Arabic 3.m.sg enclitic [h] ~ [u]

Mascaró (1996)

[h]/ V__		[u] / C__	
xtʕa=h	'his error'	*ktab-u*	'his book'
mʕa=h	'with him'	*menn-u*	'from him'
ʃafu=h	'they saw him'	*ʃaf=u*	'he saw him'

- Classical Arabic, a single morph: /hu/
- The alternation [h]~[u] cannot be derived by the regular phonology of the language synchronically in Moroccan Arabic
- This is an example of phonologically-conditioned allomorphy (PCA) (/V__ vs. /C__)

Phonological optimization: Moroccan Arabic

- xtʕa=h ‘his error’

/xtʕa=h-{h,u}/	Max	Dep	Onset	NoCoda
a. ☞ xtʕah				*
b. xtʕa.u			*!	

- ktab=u ‘his book’

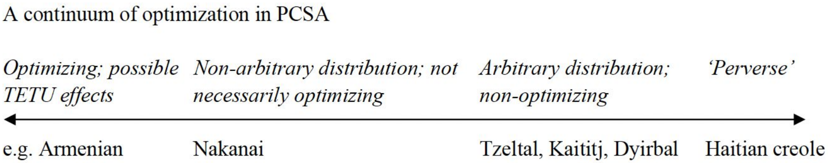
/ktab=h-{h,u}/	Max	Dep	Onset	NoCoda
a. ktabh				*
b. ☞ kta.bu			*!	

- All listed allomorphs come "for free" (no faithfulness violations)

Haitian Creole definite articles

Perverse optimization?

- very similar surface allomorphs: -a ~ -la
- (apparently) a case of ‘perverse’ allomorphy within a continuum of optimization (in Paster’s terms).



Paster (2015: 238 (26))

Questions

- Can the surface allomorphs be reduced to a single UR (so, no allomorphy)?

- If not,
  - is the distribution predictable?
    - determined in the phonology
  - Is the distribution arbitrary?
    - determined prior to the phonology (VI)

Distribution of surface allomorphs

- -la after a C-final stem

  liv-la 'the book'
- -a after vowel-final stems
  - a] → nothing else happens

    papa-a 'the father'
  - other vowels] → a glide (j, w), inserted between stem and clitic
    - Quality of the glide depends on quality of preceding vowel

      [j]: laplija 'the rain', papjeja 'the paper'

      [w]: tuwa 'the hole', batowa 'the boat'

Why ‘perverse’ (assuming 2 URs)?

- When -a is chosen instead of -la

why	[pa.pa.a]	*Ons, *OCP	instead of	*[pa.pa.la]	✓Ons, ✓OCP
	[la.pli.ja]	*Dep	instead of	*[la.pli.la]	✓Dep

- When -la is chosen instead of -a

why	[liv.la]	*NoCoda	instead of	*[li.va]	✓NoCoda

Surface allomorphy as phonology?

- A single exponence, with a floating /l/: <l>a
    - Tricky to prevent the docking of [l] when it doesn't surface
    - Tricky to account for glide insertion instead of /l/ realization

Assuming -a and -la as allomorphs

- Ordered allomorphs, constraint Priority
  - {a > la}

- Some constraints
  - Align-R: align the right edge of the stem with the right edge of a syllable
  - *C.V: constraint against the worst syllable contact

Selection of allomorphs

liv-la ‘the book’

/liv={a,la}/	R-Align	*C.V	Priority	Onset	Dep
a. ☞ liv].la			*		
b. liv].a		*!		*	
c. li.v]a	*!				

papa-a ‘the father’

/papa={a,la}/	R-Align	*C.V	Priority	Onset	Dep
a. pa.pa].la			*!		
b. ☞pa.pa].a				*	

Accounting for glide insertion

bato-wa ‘the boat’

/bato={a,la}/	Priority	Agr-Front	Agr-Round	Onset	Dep
a. ☞ ba.to].wa					*
b. ba.to].ja		*!			*
c. ba.to].a				*!	

papa-a ‘the father’, incorporating Agr constraints

/papa={a,la}/	Priority	Agr-Front	Agr-Round	Onset	Dep
a. ☞ pa.pa].a				*	
b. pa.pa].ja		*!			*
c. pa.pa].wa			*!		*

Conclusions from Haitian creole allomorphy

- Accounting for the a ~ la distribution through a single UR, tricky
  - not shown here in detail
- Assuming two URs (true allomorphy: -a, -la) and accounting for their distribution in the phonology seems hopeless at first sight
- But, after all, Haitian Creole allomorphy is not perverse at all

Galician definite articles

A tougher case

Allomorphy in the Galician definite article/3rd.acc clitic (Santiago de Compostela variety)

- The article and the pronominal clitic show the same behavior, except for one very specific environment, to be left aside for the moment.
-
    - Both generally surface as enclitics.

Default surface allomorphs

	sg	*pl*
masc	**o**	**os**
fem	**a**	**as**

Cómo=o pan	'I eat the bread'	Cf. *como*
Cómen=o pan	'they eat the bread'	Cf. *comen*
o can e o gato	'the dog and the cat'	

Restricted surface allomorphs

	sg	*pl*
masc	**lo**	**los**
fem	**la**	**las**

- The difference can be reduced to the absence/presence of /**l**/, since
  - /-o/: masculine related morph
  - /-a/: feminine related morph
  - /-s/: plural morph
- So, what needs to be accounted for is the alternation Ø~**l**.

Contexts for the l-forms

- when the preceding word (host) ends in -r or -s
- BUT -r and -s are not present on the surface (opacity)

Por comé=lo pan	'to eat the bread.'	Cf. *comér*
cóme=lo pan	'you eat the bread'	Cf. *cómes*
po=lo mar	by the sea'	Cf. *por alí*
toda=las mulleres	'all the women'	
no=lo, vo=lo	1pl=3.acc, 2pl=3.acc	

-r and -s surface elsewhere

- If ***-r*** or ***-s*** precede clitics other than 3rd person, there is no deletion.
  *Vémo**s**=**te** no espello* 'we see you in the mirror'

- The consonants ***r*** and ***s*** do not delete when followed by ***l*** in other contexts:
  *me**rl**o* 'blackbird' (word-internally)
  *se**r** **#** **l**acazán* 'to be lazy' (between words)

Positing a single UR

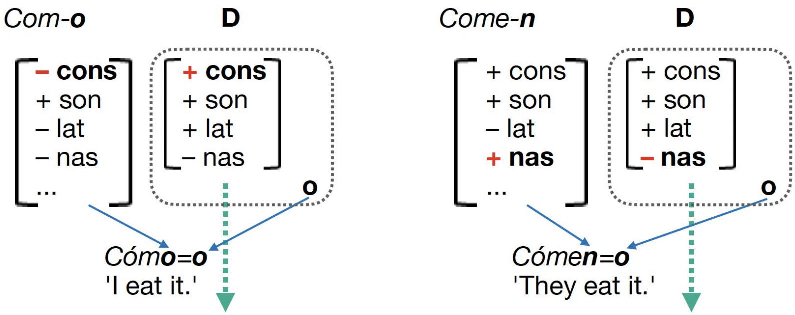
Proposals by Ulfsbjorninn (2020) and Kastner (2024)

- One single representation: floating <l>
- No deletion of -r and -s, but “blending” with the features of <l>
  - <l>: [+cons, +son, +lat, –nas]

The l~Ø alternation

- <l> cannot surface when its features are incompatible with those of the preceding segment
  - Incompatible features: [±nasal] (for Kastner), [±consonantal]
  - With compatible features, overwriting:
    - [+lateral] overwrites [–lateral]
    - [+sonorant] overwrites [–sonorant]
    - [+strident] is lost

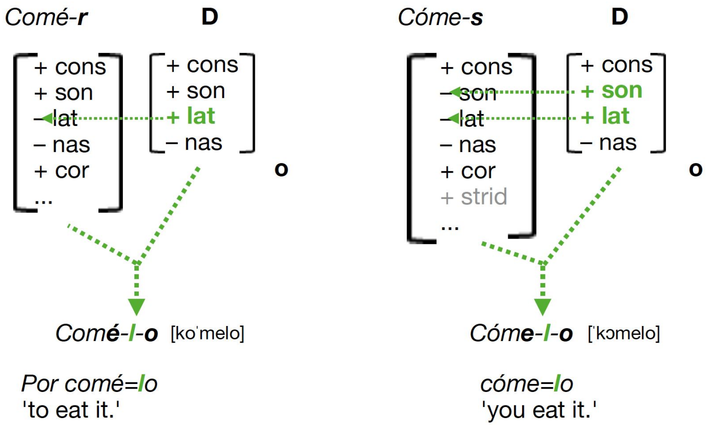
<l> failing to surface



Feature bundle not realized, because of the contradictory values of [±cons].

Feature bundle not realized, because of the contradictory values of [±nas].

<l> succeding to surface



- One would hope to find independent evidence for the behavior attributed to different features in this type of proposal.

A further complication

- Verbal forms ending in a glide surface with [n], not [l] or Ø.

coméu=o pan	's/he ate the bread'	*coméu=no pan
*coméu=**no***	s/he ate it'	*coméu=o

- It would be hard to conceive this [n] as yet another floating set of features.
- Kastner proposes a rule of n-insertion, but struggles to avoid mentioning grammatical features that will distinguish the pronoun from the article.
  - Ulfsbjorninn focuses only on the definite article.

An OT account assuming allomorphy?

- Tried by Kikuchi (2006) resorting to an Alignment constraint and Priority:
  - Align-L(MWd,σ): the left edge of a morphological word must coincide with the left edge of a syllable.

Illustration of the account

cómo=o ‘I eat’

/kómo={o,lo}/	*C.V	Align-L	Priority	Onset
a. ☞ kó.mo=.o				*
b. kó.mo=.lo			*!	

comé=lo ‘to eat it’

/komeɾ={o,lo}/	*C.V	Align-L	Priority	Onset
a. ☞ ko.mé=.lo			*	
b. ko.méɾ=.o	*!			*
c. ko.mé.ɾ=o		*!		

Complications

- But when one goes into details, the account ends up not working
  - predicts r-deletion in contexts where there isn't;
  - For 'to eat it' it wrongly predicts an output with r-deletion and selection of the wrong allomorph (*comé=o).
    - See Kastner (2024) for details

- A complete satisfactory account of the distribution of the whole set of surface allomorphs (Ø, l, n) is still awaiting.

Allomorph selection prior to the phonology?

- A possible DM-style account

Vocabulary Items:

- Obj → n /j, w__
- D → l /s, r__
- → Ø (elsewhere)

Readjustment Rule: s, r → Ø / ___l_D

Yiddish derived adjectives

Yiddish derived adjectives

aˈrum-ɪk	'surrounding'	derˈbarəm-dɪk	'merciful'
ˈraʃ-ɪk	'noisy'	ˈanderʃ-dɪk	'different'
ˈroz-ɪk	'pink'	ˈbaɪtəv-dɪk	'changeable'
tsi-ɪk	'stretchy'	’groʃə-dɪk	'money-seeking'
bɔɪg-ɪk	'flexible'	ˈfɪʃl̩-dɪk	'fish-like'
gəˈher-ɪk	'proper'	ˈglɛkər-dɪk	'bell-like'
brɛχ-ɪk	'breakable'	ˈpɛsəχ-dɪk	'for Passover'
baɪχ-ɪk	'pot-bellied'	ɔɪsərˈfokəs-dɪk	'out of focus'
jɛn-ɪk	'that'	ˈzaɪən-dɪk	'being'
aɪl-ɪk	'hurried'	ˈfrɛstl̩-dɪk	'chilly'

- -ɪk after stressed σ, -dɪk after stressless σ

The puzzle of the ɪk~dɪk distribution

- Why prefer an extra C after stressless syllables?
- Subset of cases that looks sensible
  - groʃə-dɪk, fɪʃl̩-dɪk (*əV, *C̩V)
- But the simple full generalization is just about stress, and it's not obvious why this should be so.

Claim: this helps the suffixed form be faithful in some way to the "syllabification" of the base form.

Syllabification assumptions

- Intervocalic C is syllabified as an onset: V.CV
- Post-tonic C is additionally ambisyllabic: VC̣V
  - "Coda Capture" (Kahn 1976)
  - Stress-to-Weight (SWP)
  - Using C̣ here to represent C straddling the boundary
- Compare
  - ɔtə.mən 'breathe'
  - saṃət 'velvet'

Ambisyllabicity allows for -ɪk after stress

- SWP requires stem C to be (partly) in coda
- Ambisyllabic candidate satisfies Onset
- Dep(C) and Max(C) ensure that in general, inserting or deleting consonants is banned (not shown)

jɛn-{ɪk,dɪk}	SWP	Onset	*Stressless C̣	*CC	*C̣
a. jɛ.nɪk	*!				
b. jɛn.ɪk		*!			
c. ☞ jɛṇɪk					*
d. jɛn.dɪk				*!	

Ambisyllabicity banned after stressless σ

- After stressless syllables, SWP does not compel ambisyllabicity, and V.CV syllabification is required
- Expected: incorrect [zaɪ.ə.nɪk]
- What favors [zaɪ.ən.dɪk]?

zaɪən-{ɪk,dɪk}	SWP	Onset	*Stressless C̩	*CC	*C̩
a. ☞ zaɪ.ə.nɪk					
b. zaɪ.ən.ɪk		*!			
c. zaɪ.ən̩ɪk			*!		*
d. 😿 zaɪ.ən.dɪk				*!	

Faithfulness to syllable structure

- Base: [zaɪən][*]
- Candidate [zaɪ.ə.nɪk]: coda in base, onset in candidate
- BD-Ident(syl struc) (or syllabic role; Kenstowicz 1996)

zaɪən-{ɪk,dɪk}	BD-Ident(syl struc)	SWP	Onset	*Stressless C̩	*CC	*C̩
a. zaɪ.ə.nɪk	*!					
b. zaɪ.ən.ɪk			*!			
c. zaɪ.ən̩ɪk				*!		*
d. ☞ zaɪ.ən.dɪk					*	

[*]Most "stems" of -(d)ik adjectives are straightforwardly free-standing forms. This particular example is more problematic than most; [zaɪən] is a free standing form, but its use is restricted

Summarizing PCA

Distinguishing selection from optimization

Two ways to refer to phonological contexts

- Selection at the point of insertion (morphology)
  - E.g., NOM ↔ -ka/_V, C elsewhere
- Optimal satisfaction of constraints (phonology)
  - V-ka preferred by *CC

Which one is "right"?

Theory-building from the "worst case scenario"

- Some cases of allomorphy aren't phonologically conditioned
- Some cases that are basically phonologically conditioned turn out to have exceptions (English -d/-t)
- Some cases that are basically phonologically conditioned require "clever" phonology (e.g., constraints that otherwise don't alternations)

Do we give up and call everything selection at the level of morphological insertion? (Paster, Rolle, ...)

Choosing an approach

- Different empirical predictions
  - Surface vs underlying context (Kalin 2026)
  - Generalization, errors, diachronic change
- Methodological stance
  - Even if not all allomorphy can be derived phonologically, still useful to try
  - Learn more about the language, unify analyses
  - Make predictions about preferred distributions, typological trends

**Which is right?* ⇒ *Which do learners prefer*

Wrap-up and outlook

The ingredients of a successful model

- Three general approaches
  - Prosodic boundaries
  - Cyclic derivation
  - Transderivational correspondence
- No single model captures the full range of data
- Combination of prosody and {cycles/correspondence}

The problem that phonology solves

Hypothesis from class 1: "Optimize accuracy of learning, producing, and recognizing morphemes by restricting sounds and sound combinations."

- Limited inventory of sounds, maximally distinct
- Minimize alternations (uniformity)
- Cues to morphological segmentation (prosody)

The problem that phonology solves

Cyclic alternative?

- Maybe: "Optimize rapid incremental production and parsing of complex structures" (???)
- First-pass evaluation of smaller structures as soon as available? (But: directionality, and need to re-evaluate throughout derivation)
- Cues to structural domains? (prosody)

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